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AI for Bharat Hackathon

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Team Name : Robots

Team Leader Name : Ujjwal Godara

Problem Statement : Track 1 - AI for Retail, Commerce & Market Intelligence

MandiMitra: Agentic Market Intelligence Copilot

An AI-powered multi-agent system built on Amazon Bedrock that empowers 150M+ Indian farmers with real-time mandi price intelligence, sell/hold recommendations, and negotiation support - all through conversational Hindi interface.

The Problem: India has 7,000+ APMC mandis generating massive price data. Yet 86% of small farmers sell at whatever price the local arthiya (intermediary) offers, losing 15-30% of potential value due to information asymmetry. Current tools are English-only dashboards or raw government portals that lack contextual intelligence.

Our Solution: MandiMitra uses Amazon Bedrock Agents to orchestrate 3 specialist AI agents (Price Intelligence, Sell Advisory, Negotiation Prep) that query real-time data via tool-use. Farmers ask questions in Hindi like “meri soyabean kahan bechun?” and get data-driven, actionable answers with price comparisons across mandis, optimal sell timing, and downloadable negotiation briefs.

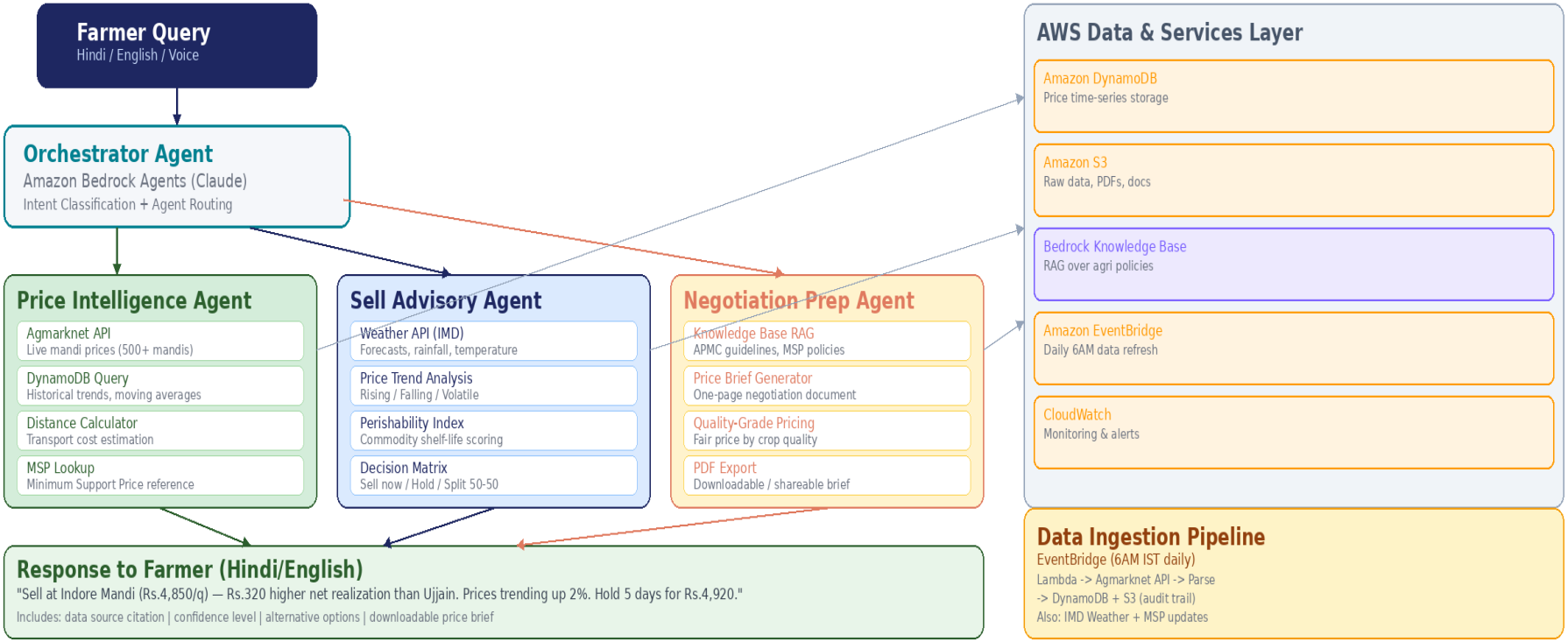
Solution Differentiation & USP

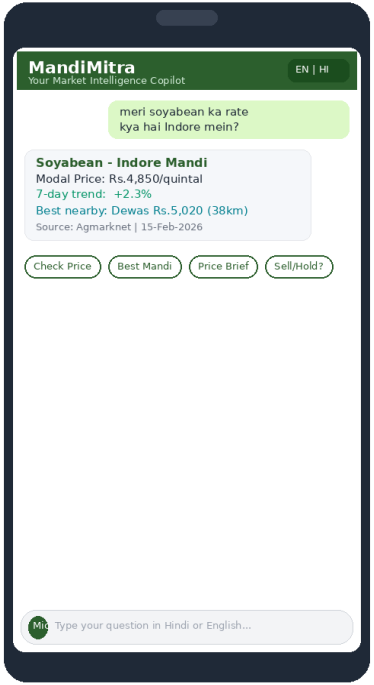
- Differentiation: Unlike Agmarknet portal, Kisan Suvidha, or eNAM (static dashboards, English-only), MandiMitra is a conversational agentic AI that reasons across multiple live data sources and provides personalized, actionable intelligence in Hindi with multi-step reasoning.
- How it solves the problem: Three specialist AI agents work together - Price Intelligence Agent compares real-time prices across 500+ mandis with transport cost adjustment, Sell Advisory Agent recommends sell/hold with weather correlation, Negotiation Prep Agent generates carry-to-mandi price briefs. All built on Amazon Bedrock with tool-use.
- USP: First agentic AI system for Indian agri-commerce combining real-time mandi data + weather-price correlation + transport-cost-adjusted net realization + multilingual conversational interface. Not a dashboard - a personal market intelligence advisor farmers interact with in Hindi.

Key Features

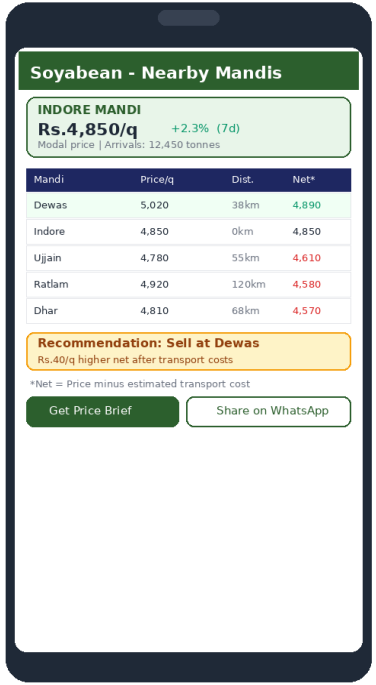
1. Real-Time Price Intelligence - Live prices for 100+ commodities across 500+ mandis with trend analysis and anomaly detection
2. Smart Sell Advisory - AI-powered sell/hold/split recommendations factoring price trends, perishability, storage costs, and weather
3. Best Mandi Finder - Transport-cost-adjusted net realization comparison across nearby mandis
4. Negotiation Price Brief - Downloadable one-page brief with MSP reference and fair price estimate
5. Hindi Conversational Interface - WhatsApp-style chat with voice input for low-literacy users
6. Weather-Market Correlation - Proactive alerts linking weather forecasts to price impacts

7. Multi-Agent Orchestration - Complex queries auto-routed across specialist agents for comprehensive answers


Farmer Query
Hindi / English / Voice
Orchestrator Agent
Amazon Bedrock Agents (Claude)
Intent Classification + Agent Routing
Price Intelligence Agent
Agmarknet API
Live mandi prices (500+ mandis)
DynamoDB Query
Historical trends, moving averages
Distance Calculator
Transport cost estimation
MSP Lookup
Minimum Support Price reference
Sell Advisory Agent
Weather API (IMD)
Forecasts, rainfall, temperature
Price Trend Analysis
Rising / Falling / Volatile
Perishability Index
Commodity shelf-life scoring
Decision Matrix
Sell now / Hold / Split 50-50
Negotiation Prep Agent
Knowledge Base RAG
APMC guidelines, MSP policies
Price Brief Generator
One-page negotiation document
Quality-Grade Pricing
Fair price by crop quality
PDF Export
Downloadable / shareable brief
AWS Data & Services Layer
Amazon DynamoDB
Price time-series storage
Amazon S3
Raw data, PDFs, docs
Bedrock Knowledge Base
RAG over agri policies
Amazon EventBridge
Daily 6AM data refresh
CloudWatch
Monitoring & alerts
Response to Farmer (Hindi/English)
"Sell at Indore Mandi (Rs.4,850/q) — Rs.320 higher net realization than Ujjain. Prices trending up 2%. Hold 5 days for Rs.4,920."
Includes: data source citation | confidence level | alternative options | downloadable price brief
Data Ingestion Pipeline
EventBridge (6AM IST daily)
Lambda -> Agmarknet API -> Parse
-> DynamoDB + S3 (audit trail)
Also: IMD Weather + MSP updates



Screen 1: Chat Interface

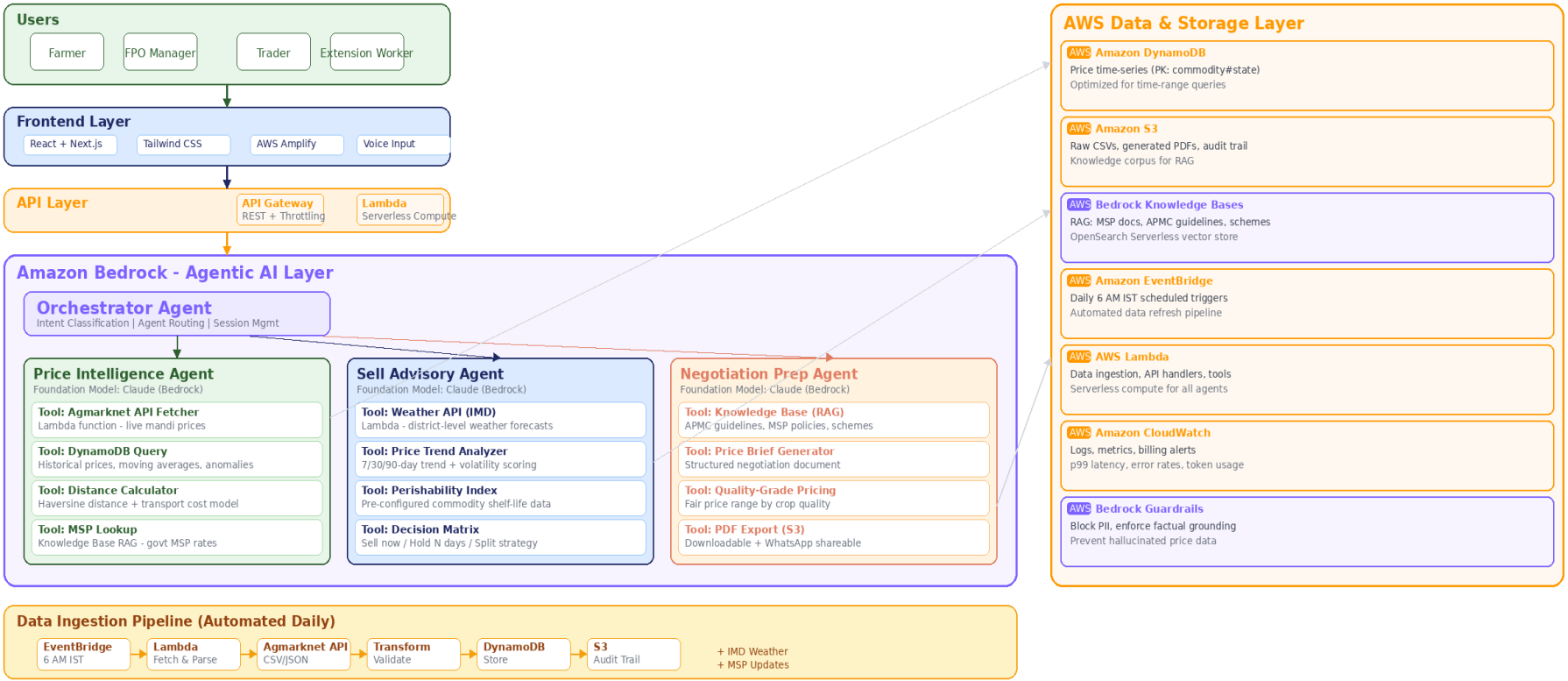


Screen 2: Mandi Price Comparison

Screen 3: Negotiation Price Brief

Key UI Elements:

- WhatsApp-style chat
  Familiar UX for rural users
- Voice input
  Hindi speech-to-text
- Quick action chips
  One-tap common queries
- Price comparison table
  Net realization ranking
- Negotiation brief
  Carry to mandi for leverage
- Share via WhatsApp
  Zero-friction distribution
- Mobile-first design
  Works on 3G/4G networks
- Large touch targets
  48x48px minimum (a11y)


Users
Farmer
FPO Manager
Trader
Extension Worker
Frontend Layer
React + Next.js
Tailwind CSS
AWS Amplify
Voice Input
API Layer
API Gateway
REST + Throttling
Lambda
Serverless Compute
Amazon Bedrock - Agentic AI Layer
Orchestrator Agent
Intent Classification | Agent Routing | Session Mgmt
Price Intelligence Agent
Foundation Model: Claude (Bedrock)
Tool: Agmarknet API Fetcher
Lambda function - live mandi prices
Tool: DynamoDB Query
Historical prices, moving averages, anomalies
Tool: Distance Calculator
Haversine distance + transport cost model
Tool: MSP Lookup
Knowledge Base RAG - govt MSP rates
Sell Advisory Agent
Foundation Model: Claude (Bedrock)
Tool: Weather API (IMD)
Lambda - district-level weather forecasts
Tool: Price Trend Analyzer
7/30/90-day trend + volatility scoring
Tool: Perishability Index
Pre-configured commodity shelf-life data
Tool: Decision Matrix
Sell now / Hold N days / Split strategy
Negotiation Prep Agent
Foundation Model: Claude (Bedrock)
Tool: Knowledge Base (RAG)
APMC guidelines, MSP policies, schemes
Tool: Price Brief Generator
Structured negotiation document
Tool: Quality-Grade Pricing
Fair price range by crop quality
Tool: PDF Export (S3)
Downloadable + WhatsApp shareable
AWS Data & Storage Layer
AWS Amazon DynamoDB
Price time-series (PK: commodity#state)
Optimized for time-range queries
AWS Amazon S3
Raw CSVs, generated PDFs, audit trail
Knowledge corpus for RAG
AWS Bedrock Knowledge Bases
RAG: MSP docs, APMC guidelines, schemes
OpenSearch Serverless vector store
AWS Amazon EventBridge
Daily 6 AM IST scheduled triggers
Automated data refresh pipeline
AWS AWS Lambda
Data ingestion, API handlers, tools
Serverless compute for all agents
AWS Amazon CloudWatch
Logs, metrics, billing alerts
p99 latency, error rates, token usage
AWS Bedrock Guardrails
Block PII, enforce factual grounding
Prevent hallucinated price data
Data Ingestion Pipeline (Automated Daily)
EventBridge
6 AM IST
Lambda
Fetch & Parse
Agmarknet API
CSV/JSON
Transform
Validate
DynamoDB
Store
S3
Audit Trail
+ IMD Weather
+ MSP Updates

Technology Stack

AI/ML Layer: Amazon Bedrock (Claude Sonnet/Haiku for reasoning), Amazon Bedrock Agents (multi-agent orchestration with tool use), Amazon Bedrock Knowledge Bases (RAG over agricultural policy documents), Amazon Titan Embeddings (vector search), Bedrock Guardrails (factual grounding, no PII)

Data Layer: Amazon DynamoDB (price time-series, optimized GSIs), Amazon S3 (raw data, PDFs, knowledge corpus), Amazon OpenSearch Serverless (vector store for Knowledge Bases)

Compute & Integration: AWS Lambda (serverless functions), Amazon EventBridge (daily data refresh scheduler), Amazon API Gateway (REST API with rate limiting & throttling)

Frontend: React 18 + Next.js 14 (SSR), Tailwind CSS (mobile-first), AWS Amplify Hosting (CI/CD)

Dev Tools: Kiro IDE (requirements & design generation), GitHub (version control), Amazon CloudWatch (monitoring, metrics, billing alerts)

External Data APIs: Agmarknet (data.gov.in - mandi prices), IMD (India Meteorological Department - weather forecasts), Ministry of Agriculture (MSP reference rates)

Estimated Monthly Cost (1,000 DAU MVP)

Amazon Bedrock (Claude Haiku - 50K queries x 1K tokens avg): ~Rs.4,000/month

DynamoDB (on-demand, 5GB storage, 500K reads): ~Rs.800/month

Bedrock Knowledge Base (OpenSearch Serverless): ~Rs.3,000/month

Lambda (100K invocations, 256MB, avg 3s): ~Rs.200 | S3 (10GB): ~Rs.100 | API Gateway: ~Rs.300

CloudWatch (logs + metrics): ~Rs.200/month

TOTAL ESTIMATED: ~Rs.8,600/month (~$100/month) | Scales to ~Rs.50,000 at 10,000 DAU

Note: 100% serverless architecture with pay-per-use pricing. No fixed infrastructure costs.

Impact, Scalability & Roadmap

TARGET IMPACT: 150M+ small farming households lose 15-30% of crop value to information asymmetry. MandiMitra democratizes market intelligence, potentially recovering Rs.50,000-1,00,000 per farmer annually.

SCALABILITY: 100% serverless (Lambda + DynamoDB). Adding new mandis, commodities, and languages requires only data pipeline config - zero architectural changes. Auto-scales to millions of users.

BUSINESS MODEL: Free for individual farmers (funded by FPO/institutional subscriptions). Premium tier for FPOs, aggregators, commodity traders with bulk analytics and API access.

GOVT ALIGNMENT: Complements eNAM (digital mandi), PM-KISAN, and Doubling Farmers Income mission. Can integrate with state agriculture department advisory systems.

EXPANSION ROADMAP: WhatsApp integration (zero-download reach) | Crop photo quality grading | eNAM direct trading | Multi-language rollout (Tamil, Telugu, Marathi, Gujarati, Punjabi) | Satellite data (NDVI) for yield-adjusted forecasting

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Thank You